
Q1.

Given that A is a Boolean variable, evaluate and then simplify the following expressions: [Storey, Electronics: A Systems Approach, 5th Ed., Pearson]

- $A \cdot 1$
- $A \cdot \overline{A}$
- $1 + A$
- $A + \overline{A}$
- $1 \cdot 0$
- $1 + 0$

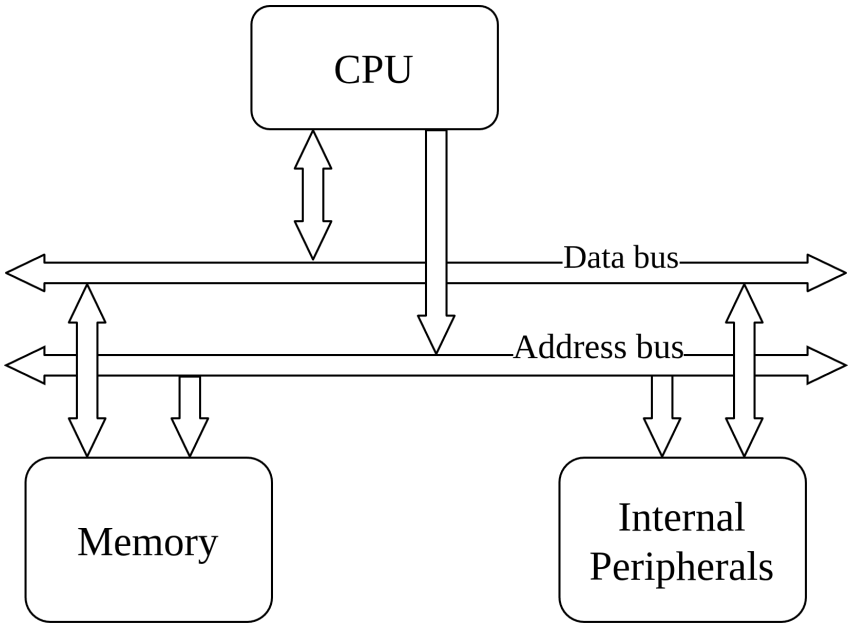
Q2.

You want to change the setting of a register in one of the internal peripherals within your ATMega32u2.

(a) First you want to change just a single bit in the register. Explain why you cannot simply write to a single bit of a register.

(b) How many bits are read/modified by default by each operation within the ATMega32u2?

(c) The register that you are wanting to modify is at address `0x0024`, and its current value is `0b1100 1000`. On the diagram below, show signals present on the address, data, and control buses, when the `READ` operation is taking place.



(d) Having read the byte, you want to force the LSB to be equal to 1. What logic operation will force a bit to be equal to 1? (*Hint: Looking at the tools in question 1, or within L4 may help*)

(e) In pseudocode or C, what operation will result in the byte being modified to set the LSB to 1, while leaving the rest of the byte unmodified?

(f) Within which part of the computer system will this operation be performed, and how will the register be updated with the new value?

(g) Extra for experts/curious only: What function does the above operation perform in the ATMega32u2? (*Hint: See section 15.9.1, p102 of the ATMega32u2 datasheet.*)

Q3.

Draw truth tables for each of the following logical functions of 3 Boolean variables:

(a) $z = f(a, b, c) = a + \overline{b} + \overline{c}$

(b) $z = f(a, b, c) = \overline{a}(b + \overline{c})$

(c) $z = f(a, b, c) = \overline{a}b\overline{c} + \overline{a}\overline{b}\overline{c} + \overline{a}bc$

(d) $g = f(p, t, u) = u + p\overline{t}$

Q4.

Apply the rules of Boolean algebra to simplify the expression:
$Q = \left(A \cdot \overline{B} \cdot C\right) + \left(A \cdot \left(\overline{B} \cdot \overline{C}\right)\right)$.

Q5.

Use truth tables to prove De Morgan's laws: $\overline{A} \cdot \overline{B} = \overline{A + B}$ and $\overline{A} + \overline{B} = \overline{A \cdot B}$.

Q6.

Using a Karnaugh Map, simplify the following expression:

$$f(a,b,c,d) = \overline{a}\overline{b}\overline{c}\overline{d} + \overline{a}\overline{b}c\overline{d} + \overline{a}\overline{b}\overline{c}d + \overline{a}\overline{b}cd + \overline{a}bc\overline{d} + a\overline{b}c\overline{d} + abc\overline{d}$$

Q7.

Lecture 5 (CPU) tutorial questions
(*Note, we won't have covered these for the tutorials 11/8*)

(a) Write an expression for a full adder's C_{out} in terms of the inputs A, B, and C_{in}.

(b) Draw a circuit using AND, OR, and NOT logic gates that implements your expression for C_{out}.

(c) Consider a comparator that operates on two two-bit numbers: $A = A_1A_0$, and $B = B_1B_0$:

  (a) Write a logic expression for the *gt* output of this comparator. Your expression should use A_0, A_1, B_0, and B_1 as inputs.

  (b) Draw a logic gate diagram based on your logic expression.

  (c) Imagine how incredibly complex an 8-bit comparator would be using the same design process.

Q8.

For **Q3** (a) & (b), represent each of the functions using only `NAND` gates.

END OF QUESTIONS